GURONG PINOY

TEAM VYGOTSKY

REFRESHER

GENED Mathematics

GURONG PINOY

LET & CSE

#1 ONLINE REVIEW AUTHORITY

Gurong Pinoy

PRAYER

Dear Lord, I come to you to ask for your guidance and direction in this study session. I ask that the holy spirit fill me with strength, creativity and understanding to get through my studies without difficulty or sin. Help me hold my focus and attention. Help me to retain all that I learned. Please make my mind sharp and keep distractions at bay. If any part of my studying is weak or lacking in some way, let it be revealed so that I may correct it. Thank you for this opportunity to learn. Amen.

MATH ka lang...
Magiging
LPT
AKO!!

1. One-way fare from Manila to Cebu is ₱3,800. How many round-trip tickets can be purchased with the amount of ₱91,200?

a. 8
b. 14
c. 10
d. 12

₱3,800 = òne-way fare

₱3,800 X 2 = ₱7,600 - cost of 1 round-trip ticket

₱91,200 / ₱7,600 = 12 round trip tickets

2. A water tank contains 18 liters when it is 20% full. How many liters does it contain when 50% full?

a. 60
b. 30
c. 58
d. 45

P = 18

R = 20% or 0.20

B = ?

B = P/R

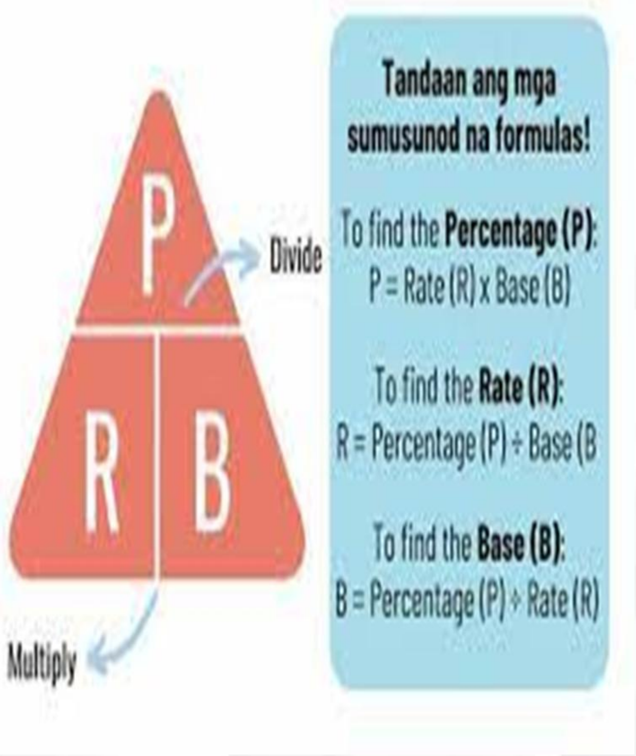
B = 18/0.2

B = 90 liters when full

When 50% full:

= 90 x 0.5

= 45 liters



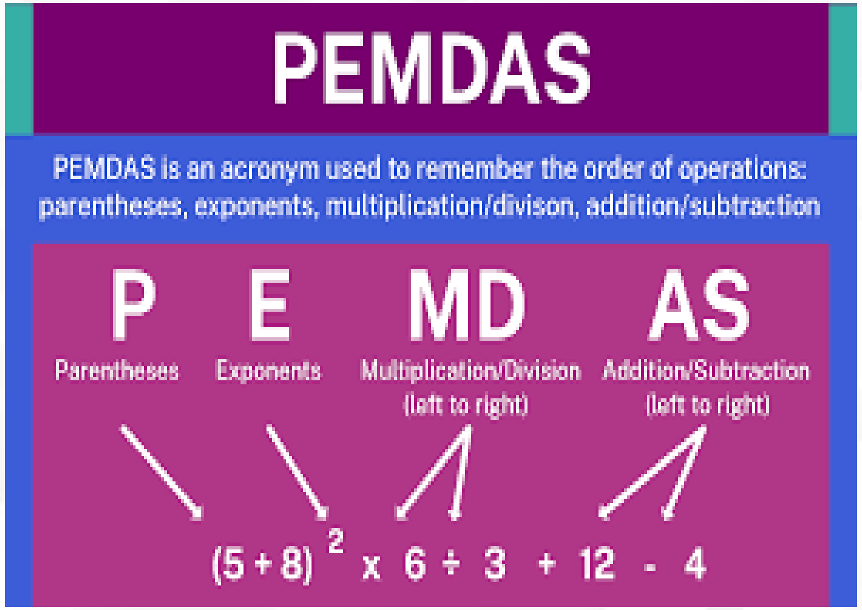
3. Which of the following has the greatest value?

a. $3 + 3^2 + (3 + 3)^2$

b. 3^3

c. $(3 + 3 + 3)^2$

d. $[(3 + 3)^2]^3$



$3 + 3^2 + (3 + 3)^2 \quad = 3 + 9 + 36 \quad = 48$

$3^3 \quad = 27$

$(3 + 3 + 3)^2 \quad = 9^2 \quad = 81$

$[(3 + 3)^2]^3 \quad = \{(6)^2\}^3 \quad = (36)^3 \quad = 46656$

4. The average of five counting numbers is 20. What is the highest possible value that one of the numbers can have?

a. 20
b. 30
c. 40
d. 90

Let: 1, 2, 3, 4 - the first four numbers
N - the highest number
20 - average of the five numbers

Formula of Average

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$$\text{average} = \frac{\text{sum of values}}{\text{number of values}}$$

$$20 = \frac{1+2+3+4+N}{5}$$

$$5\left(20 = \frac{1+2+3+4+N}{5}\right)5$$

$$100 = 10 + N$$

$$100 - 10 = 10 - 10 + N$$

$$90 = N$$

5. Thirty students had an average of 75, while 20 students had an average of 90. What is the average of the entire group?

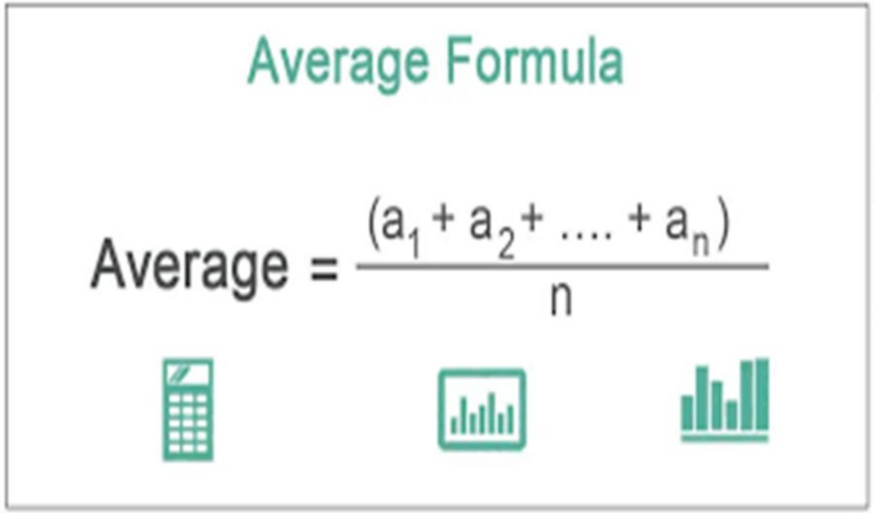
a. 81
b. 82
c. 82.5
d. 81.5

$$\text{Ave} = \frac{(30)(75)+(20)(90)}{50}$$

$$\text{Ave} = \frac{2{,}250+1{,}800}{50}$$

$$\text{Ave} = \frac{4{,}050}{50}$$

Ave = 81



6. How much greater is the sum of the first 100 counting numbers than the sum of the first 50 counting numbers?

a. 110
b. 3775 ✓
c. 3177
d. 1200

$$S_{n(50)} = \left(\frac{50}{2}\right)(1 + 50)$$

$$S_{n(50)} = 25 \times 51$$

$$S_{n(50)} = 1{,}275$$

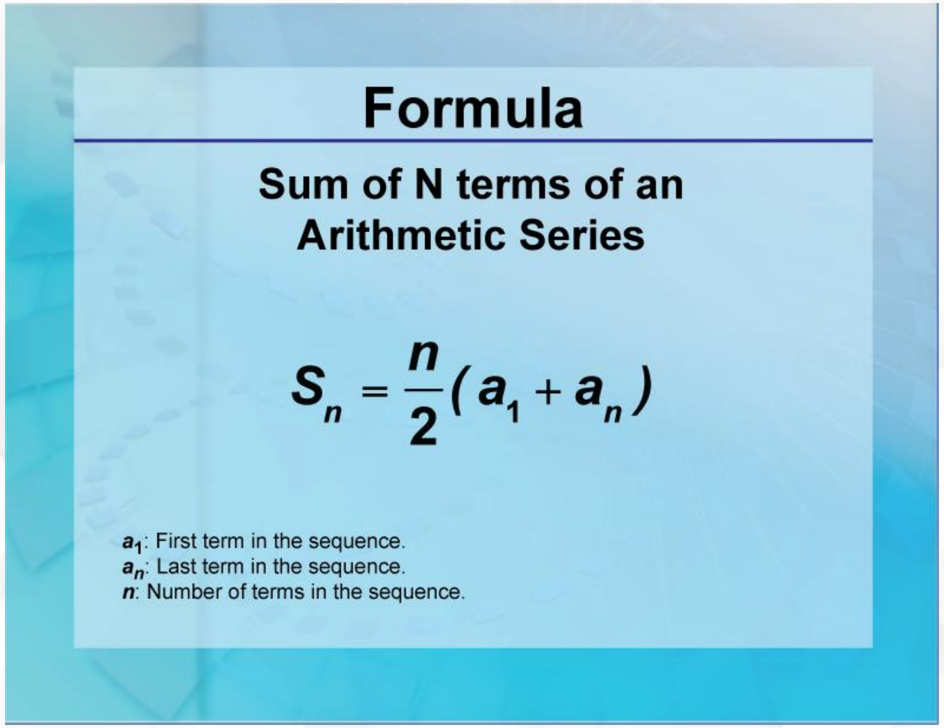
$$S_{n(100)} = \left(\frac{100}{2}\right)(1 + 100)$$

$$S_{n(100)} = 50 \times 101$$

$$S_{n(100)} = 5{,}050$$

$$S_{n(100)} - S_{n(50)} = 5{,}050 - 1{,}275$$

$$S_{n(100)} - S_{n(50)} = 3{,}775$$



7. If a man uses 20 gallons of gasoline to travel 350 km in his car, how many gallons will he need for a 750 km trip?

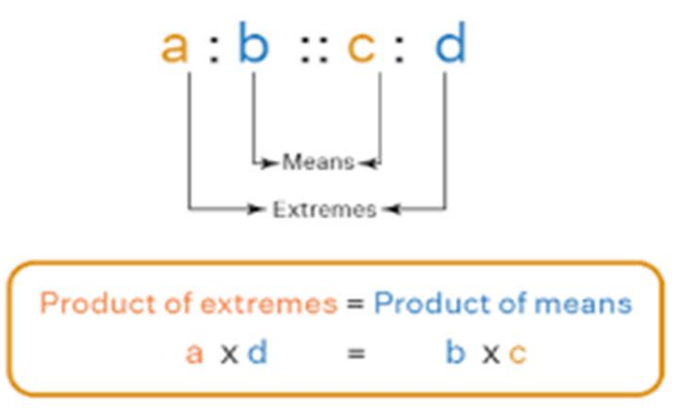
a. 42.86
b. 52.86
c. 62.86
d. 72.86

$$20 : 350 = X : 750$$

$$15{,}000 = 350X$$

$$15{,}000/350 = 350X/350$$

$$X = 42.857 \text{ or } 42.86$$



8. Which of the following is the factorization of the binomial $x^2 - 4$?

a. $(x+4)(x-2)$

b. $(x+2)(x+2)$

c. $(x-2)(x-2)$

d. $(x-2)(x+2)$

Difference of Squares Formula.

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DIFFERENCE OF SQUARES FORMULA

$$a^2 - b^2 = (a + b)(a - b)$$

$$x^2 - 4$$

$$= x^2 - 2^2$$

$$= (x+2)(x-2)$$

Check using the FOIL method

$$= (x-2)(x+2)$$

$$= (x^2 + 2x - 2X - 4)$$

$$= x^2 - 4$$

9. The vertex of an isosceles triangle is 20 degrees. What is the measure of one of the base angles?

a. 150 degrees

b. 75 degrees

c. 80 degrees

d. 90 degrees

Let: x be measure of one base angle

$$x + x + 20 = 180$$

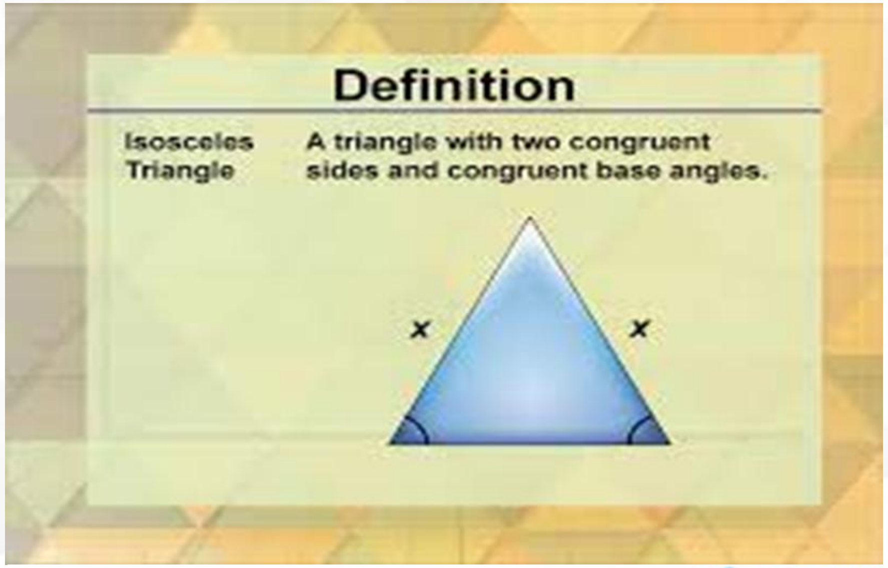
$$2x + 20 = 180$$

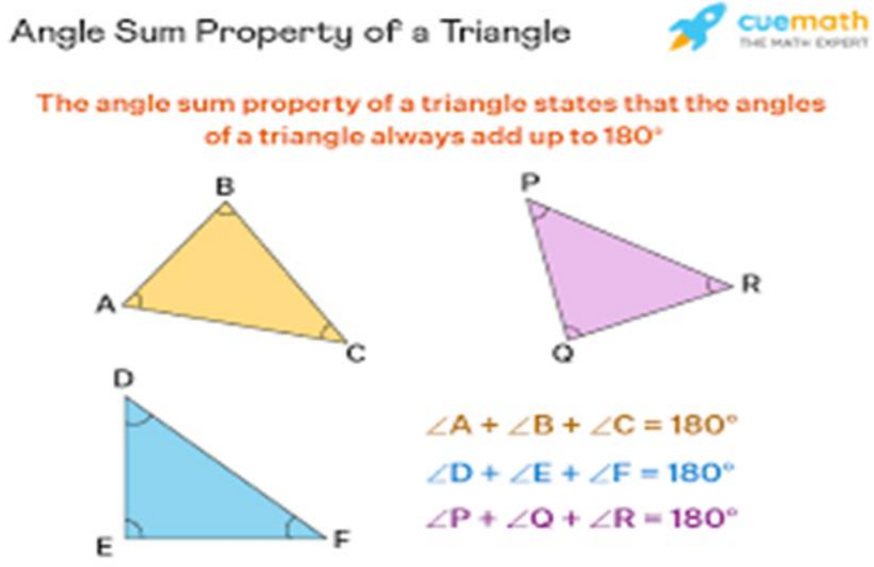
$$2x = 180 - 20$$

$$2x = 160$$

$$2x/2 = 160/2$$

$$x = 80$$





10. Only 26% of 50 students passed the exam. How many students failed the test?

a. 37
b. 35
c. 33
d. 39

B = 50 total no. of students

R = 26% or 0.26 passed

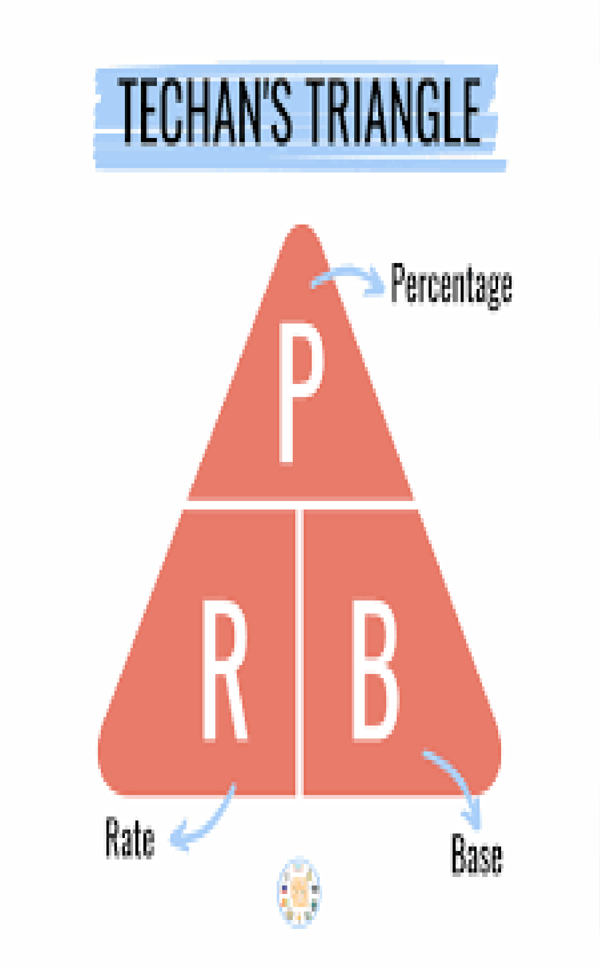
P = ?

P = R x B

P = 0.26 X 50

P = 13 – no of students who passed

50 – 13 = 37



11. How much commission does a salesman get if he was able to sell ₱35,000 worth of jewelry and he is given a 15% of it?

a. ₱5,300
b. ₱5,325
c. ₱5,250
d. ₱5,425

B = ₱35,000

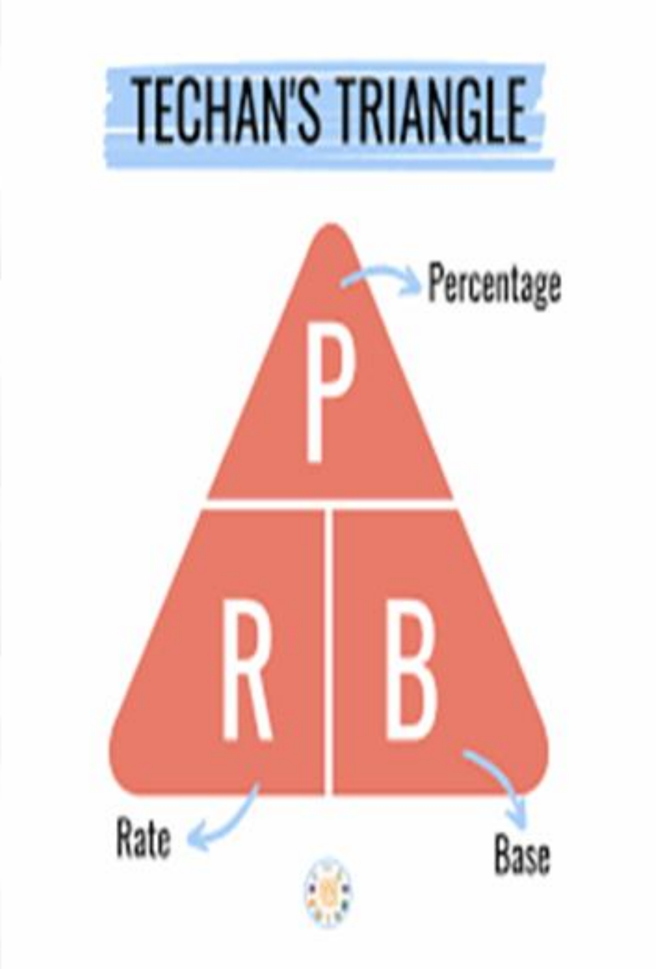
R = 15% or 0.15

P = ?

P = R x B

P = 0.15 x ₱35,000

P = ₱5,250



12. How much should a man deposit in a bank at 6% simple interest rate if he wants to earn ₱1,200 a year?

a. ₱10,000

b. ₱30,000

c. ₱15,000

d. ₱20,000

I = ₱1,200

R = 6% or 0.06

T = 1 year

P = ?

I = PRT

$$1{,}200 = P(0.06)(1)$$

$$1{,}200 = 0.06P$$

$$1{,}200/0.06 = 0.06P/0.06$$

$$20{,}000 = P$$

Simple Interest = P × r × t

- P → Principal
- r → Interest Rate
- t → Time in Years

13. Julius has ₱15,000 in the bank. After a year, he expects it to earn ₱750. What is the rate of interest?

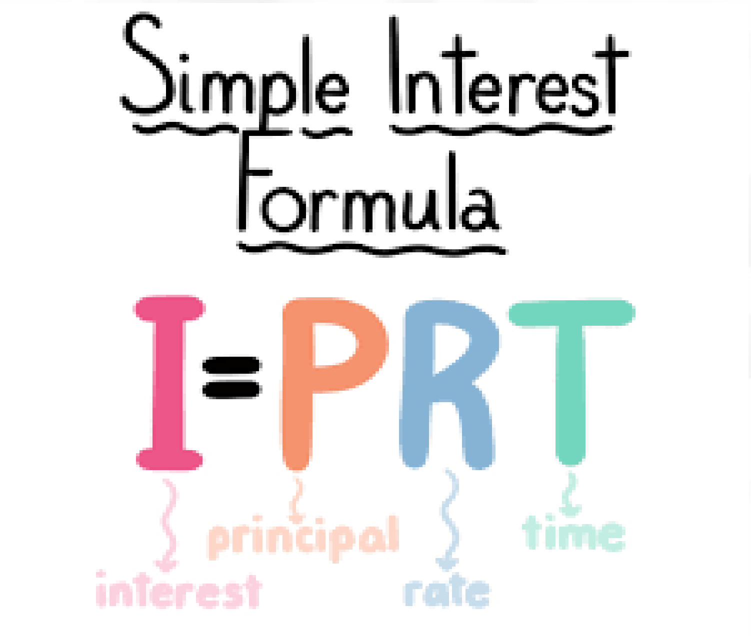
a. 2%
b. 4%
c. 5%
d. 6%

P = P15,000
T = 1 year
I = P750
R = ?

I = PRT
750 = (15,000) x R x 1
750 = 15,000R

750/15,000 = 15,000R/15,000

0.05 = R or R = 5%



14. In a playground for kindergarten kids, 18 children are riding tricycles or bicycles. If there are 43 wheels in all, how many tricycles are there?

a. 8
b. 9
c. 7
d. 11

$T + B = 18$ ------ eq 1

$3T + 2B = 43$ ------- eq 2

Multiply eq 1 by -2 then add to eq 2

$3T + 2B = 43$ ------- eq 2

$-2T - 2B = -36$

$T = 7$ No. of tricycles

$18 – 7 = 11$ No. of bicycles

To check:

$3T + 2B = 43$

$3(7) + 2(11) = 43$

$21 + 22 = 43$

$43 = 43$

15. How much will P25,000 amount to at 5% simple interest rate after 2 years?

a. P27,500
b. P28,500
c. P2,500
d. P27,000

P = P25,000
r = 5% or 0.05
t = 2 years

A = ?

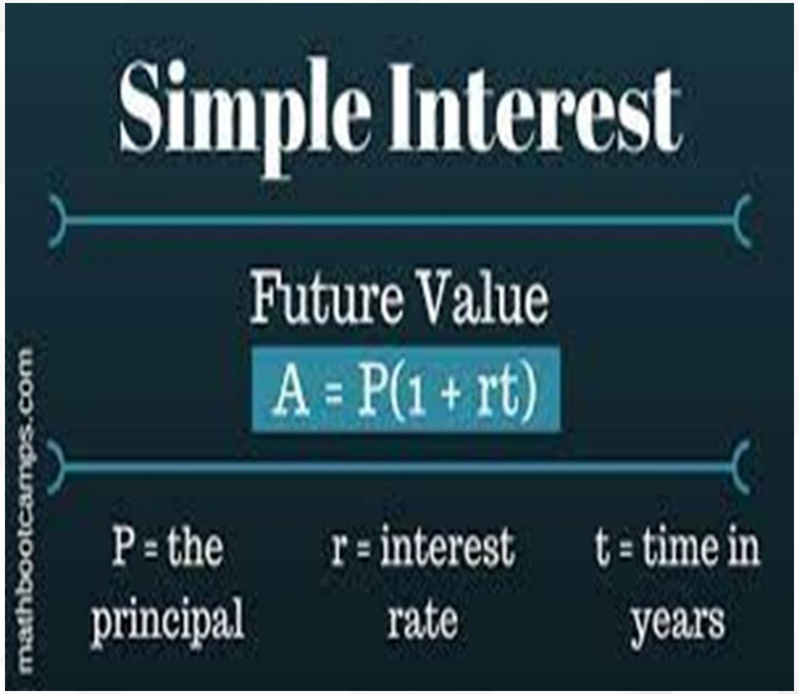
$A = P(1 + rt)$

$A = 25,000[1 + (0.05)(2)]$

$A = 25,000(1 + 0.1)$

$A = 25,000 + 2500$

A = P27,500



16. Which has the most volume?

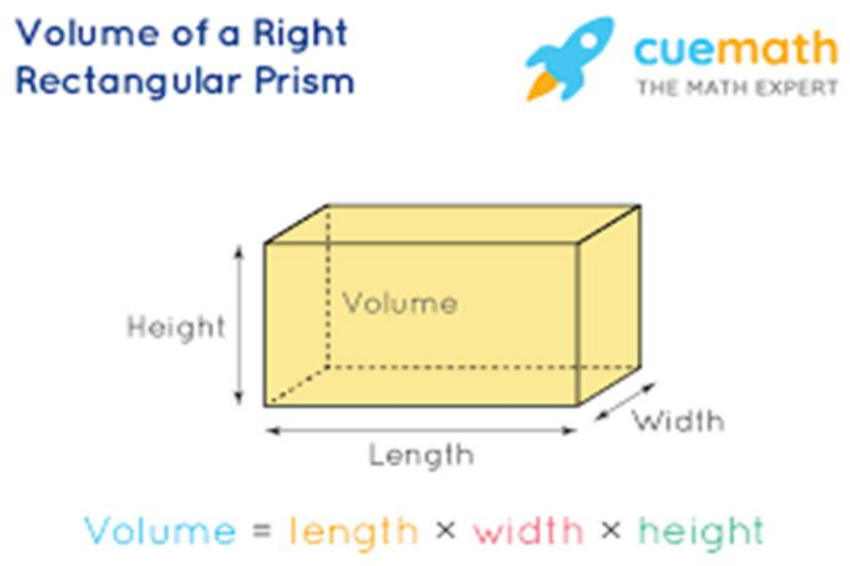
a. A box 10 meters high, 6 meters wide, and 14 meters long.
b. A box 1 meter high, 60 meters wide, and 14 meters long.
c. A box 9 meters high, 7 meters wide, and 14 meters long.
d. A box 11 meters high, 8 meters wide, and 7 meters long.

a. 10m x 6m x 14m = 840 cubic meters

b. 1m x 60m x 14m = 840 cubic meters

c. 9m x 7m x 14m = 882 cubic meters

d. 9m x 7m x 14m = 616 cubic meters



17. The edges of a rectangular solid have these measures: 1.5 ft by 11/2 ft by 3 inches. What is its volume in cubic inches?

a. 324
b. 225
c. 972
d. 45

1 ft. = 12 in.

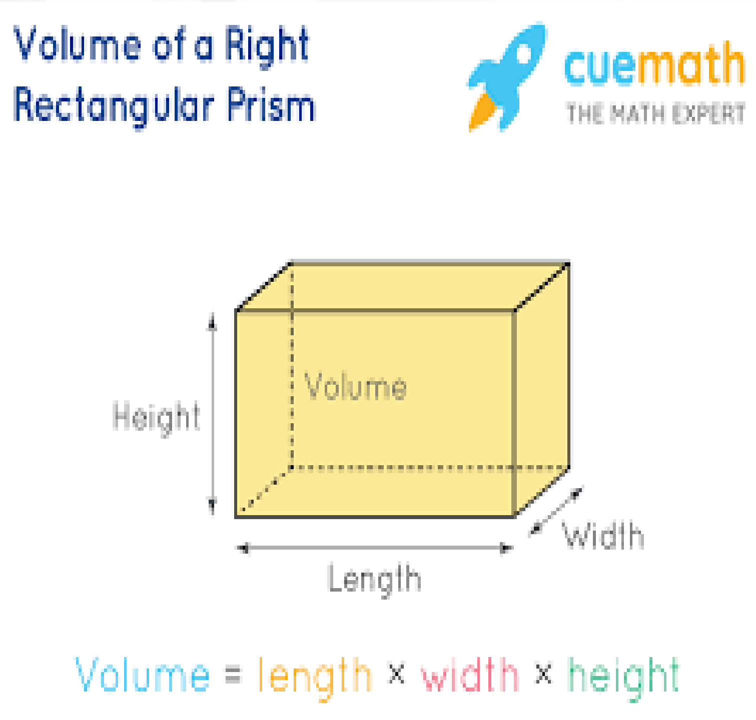
1.5 ft. X 12 in./ft = 18

11/2 ft. X 12 in./ft = 18

V = L x W x H

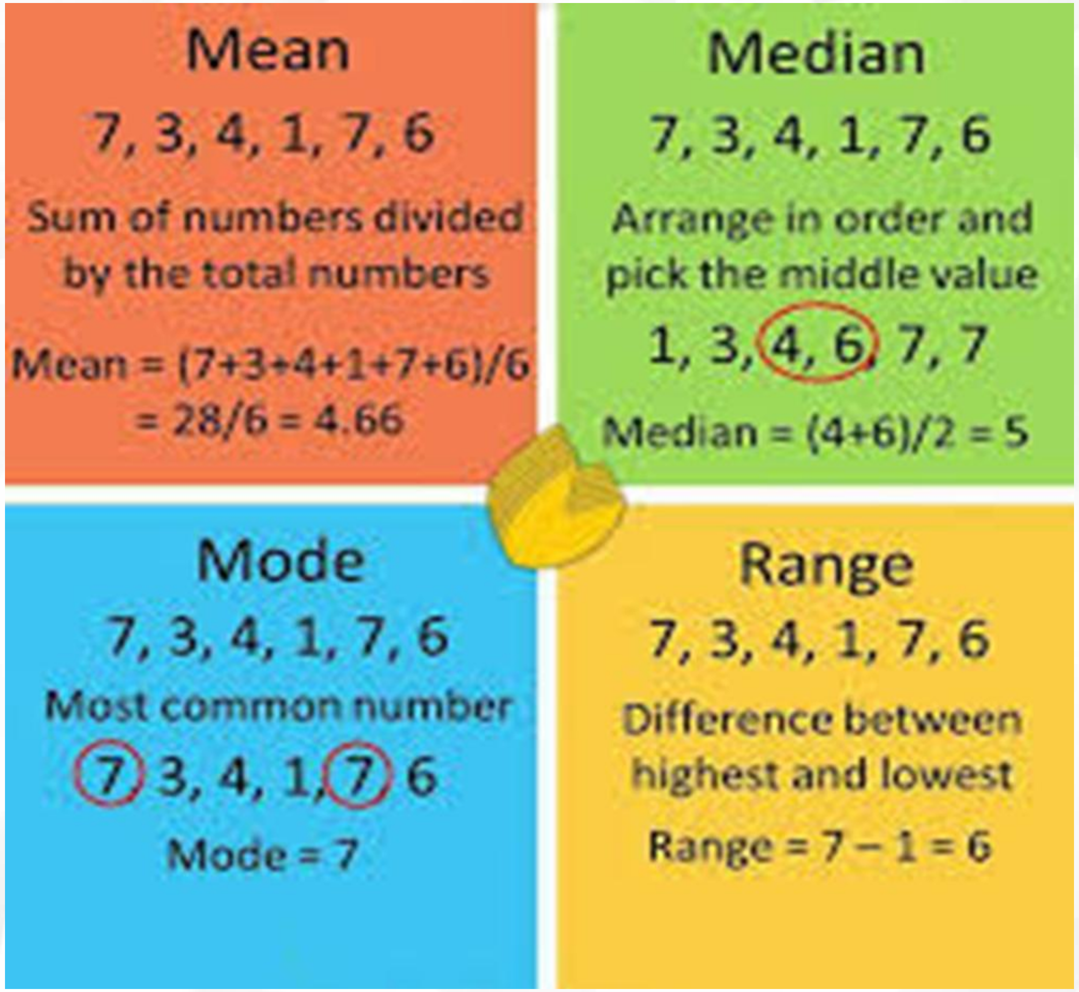
V = 18 x 18 x 3

V = 972 cu. in.



18. If the score of 10 students are: 76, 80, 75, 83, 80, 79, 85, 80, 88, 90, the mode is:

a. 85
b. 79
c. 88
d. 80



19. Find the median of the following numbers 8, 5, 7, 5, 9, 9, 1, 8, and 10

a. 5
b. 7
c. 8
d. 9

If n is odd

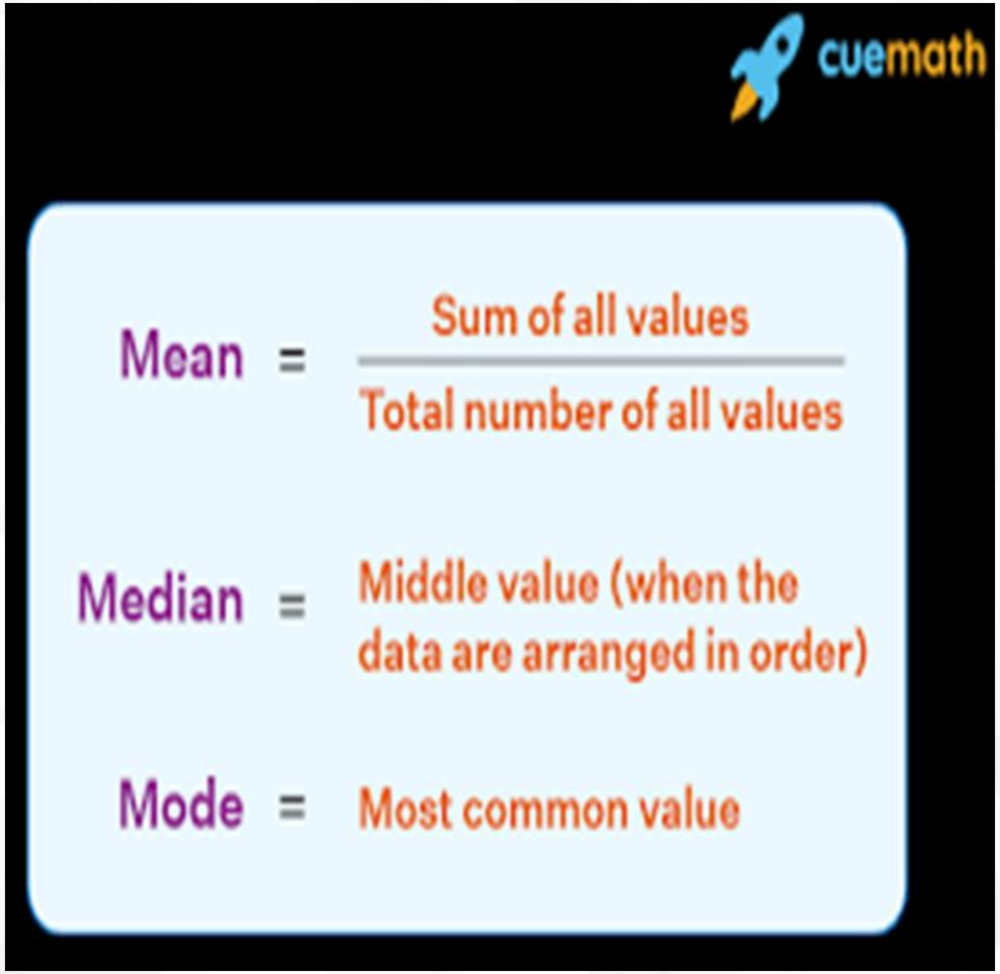
1, 5, 5, 7, 8, 8, 9, 9, 10

Median = 8

If n is even

1, 3, 5, 5, 7, 8, 8, 9, 9, 10

Median = (7 + 8)/2 = 7.5



20. Ruben’s grades are 88, 90, 97, 90, 91 and 86. What is the grade that he should aim for in the 7th subject if he has to have an average of 91?

a. 97

b. 95

c. 92

d. 89

$$91 = \frac{88+90+97+90+91+86+N}{7}$$

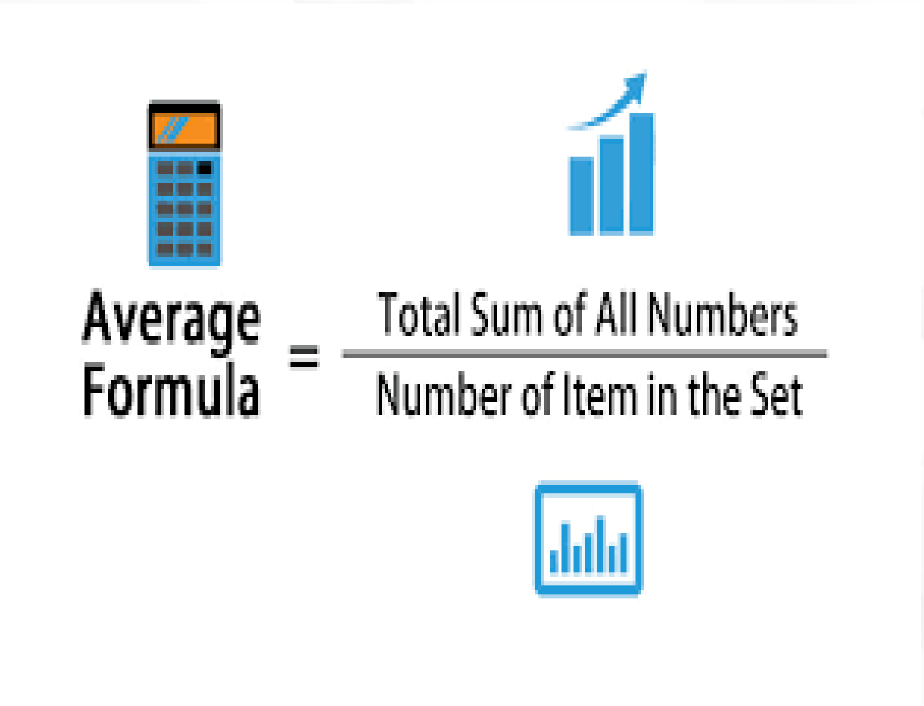
$$91 = \frac{542+N}{7}$$

$$7\left(91 = \frac{542+N}{7}\right)7$$

$$637 = 542 + N$$

$$637 - 542 = N$$

$$95 = N$$



21. If a certain job can be finished by 18 workers in 26 days, how many workers are needed to finish the job in 12 days?

a. 24
b. 30
c. 39
d. 45

$$18:26 = x:12$$

$$18\ (26) = 12\ (x)$$

$$468 = 12x$$

$$468/12 = 12x/12$$

$$\mathbf{39 = x}$$

$$18 \times 26 = 39 \times 12$$

$$468 = 468$$

22. Three men are paid P315 to do a certain job. The sum is to be divided among them in the ratio of 4:5:6. How much does each receive?

a. 84, 104, & 127
b. 84, 105, & 126
c. 83, 106, & 126
d. 83, 105, & 127

$$4x + 5x + 6x = 315$$
$$15x = 315$$
$$15x/15 = 315/15$$
$$x = 21$$

$$4x = 4(21) = 84$$
$$5x = 5(21) = 105$$
$$6x = 6(21) = 126$$

23. The product of two whole numbers is 36, and their ratio is 1:4. Which of these is the smaller number?

a. 9
b. 3
c. 2
d. 12

Since the product of two numbers is 36, look for the factors of 36 that can satisfy the ratio 1:4

Factors of 36 :

1 x 36 = 1 : 36

2 x 18 = 1 : 9

3 x 12 = 1 : 4

4 x 9 = 4: 9

6 x 6 = 1 : 1

24. Three times the first of the three consecutive integers is three more than twice the third. Find the third integer.

a. 9
b. 13
c. 11
d. 15

Let x – first integer

$x + 1$ – 2^{nd} integer

$x + 2$ – 3^{rd} integer

$3x = 2(x + 2) + 3$

$3x = 2x + 4 + 3$

$3x = 2x + 7$

$3x - 2x = 7$

$x = 7$ – 1^{st} integer

$x + 2 = 9$ – 3^{rd} integer

To check:

$3 \times 7 = [2 \times 9] + 3$

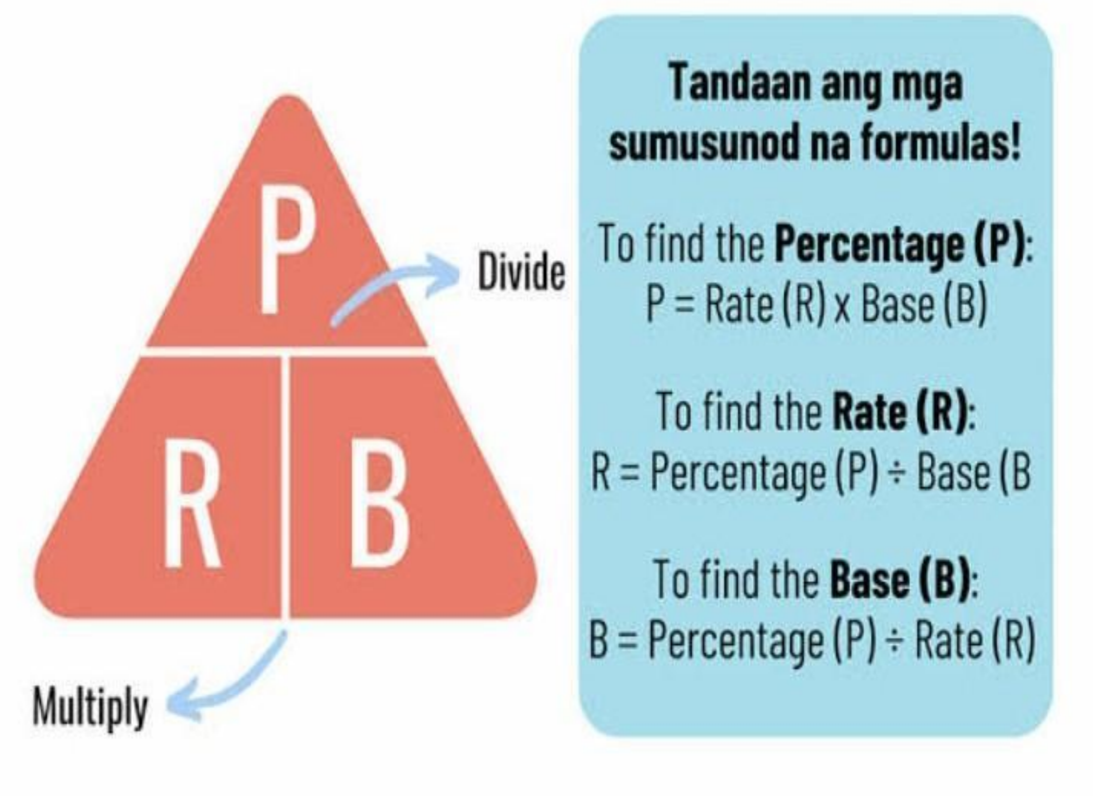
$21 = 18 + 3$

$21 = 21$

25. Julio's salary is ₱8,500 per month. If it is increased by 12%, what is his new salary per month?

a. ₱9,320
b. ₱9,420
c. ₱9,520
d. ₱9,550

= ₱8,500 x 0.12
= ₱1,020
= ₱8,500 + ₱1,020
= ₱9,520
or
= ₱8,500 x 1.12
= ₱9,520



26. An article of clothing is on sale at 15% off the marked price. If the marked price is 1,050, what is the sale price?

a. 132.15
b. 892.5
c. 157.5
d. 787.5

Marked price = 1050
Discount rate = 15%
Selling Price = ?

P = R x B

P = 0.15 x 1050

P = 157.5 - Discount

SP = LP - D

SP = 1050 – 157.5

SP = 892.5

27. An elevator can carry a maximum load of 605 kg. How many passengers weighing 50.5 kg each can the elevator load?

a. 10
b. 11
c. 12
d. 13

605/50.5 = 11.98

11 persons only

28. Of the 50 students enrolled in the subject – Trigonometry, 90% took the final examination at the end of the term. Two-thirds of those who took the final exam passed. How many students passed the exam?

a. 33
b. 45
c. 34
d. 30

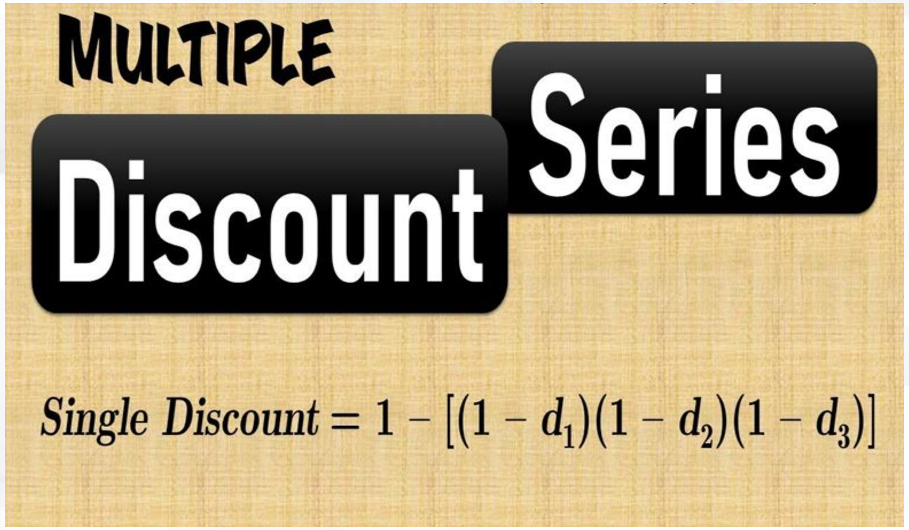
$50 \times 0.9 = 45 \longrightarrow$ Took the final exam in Trigonometry

$45 \times 2/3 = 30 \longrightarrow$ Passed the exam

29. Of the following discounts, which is equal to the series discount 30%, 20%, 10%

a. 40%
b. 46.9%
c. 48%
d. 49.6%

$$1 - d_1 = 1 - 0.3 = 0.7$$
$$1 - d_2 = 1 - 0.2 = 0.8$$
$$1 - d_3 = 1 - 0.1 = 0.9$$



$$SD = 1 - (0.7)(0.8)(0.9)$$
$$SD = 1 - 0.504$$
$$SD = 0.496 \text{ or } 49.6\%$$

30. How many times will the digit 7 appear between 1 to 100?

a. 9
b. 20
c. 19
d. 11

Whole Numbers 1 to 100

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100


You are
Stronger
than what you
Think